YT DOWNLOADER

README & USER MANUAL

Version 1.0 · Python 3.11+ · yt-dlp Engine

A modern, full-featured YouTube downloader with dark-mode GUI

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1. Overview

YT Downloader is a **modern, full-featured desktop application** for downloading YouTube videos and playlists. Built on **yt-dlp** — the most actively maintained YouTube download library — it provides a polished dark-mode GUI so you never need to touch a command line.

The application runs entirely on your machine. No data is sent to any external server except to YouTube/Google to fetch the video stream.

Key Features

- **Video & Audio formats:** MP4, WebM, MP3, M4A, Opus, WAV
- **Resolution control:** 144p through 4K, or Best Available
- **Playlist support:** download entire playlists or select individual videos
- **Real-time progress:** speed, ETA, downloaded vs. total size
- **Pause / Resume / Cancel:** full download lifecycle control
- **Thumbnail preview:** shown before download starts
- **Drag-and-drop:** drop a URL directly onto the window
- **Download history:** record of past downloads, persisted across sessions
- **FFmpeg integration:** auto-merges streams, embeds metadata and cover art
- **Dark / Light mode:** toggleable, remembers your preference
- **Persistent settings:** saved to ~/.ytdownloader/config.json

Technical Stack

Component	Technology
GUI Framework	CustomTkinter 5.2+ (modern Tkinter wrapper)
Download Engine	yt-dlp (latest — imported in ONE file only)
Post-processing	FFmpeg (must be installed separately on your system)
Thumbnails	Pillow + requests (HTTP fetch → PIL Image → CTkImage)
Threading	Python threading.Thread (daemon threads)
Config storage	JSON file at ~/.ytdownloader/config.json
Drag-and-drop	tkinterdnd2 (optional, graceful fallback if absent)
Python version	3.11 or newer required

2. Installation

2.1 Prerequisites

Python 3.11+ must be installed and on your system PATH. Check with: `python --version`

FFmpeg must be installed separately and available on PATH. It handles stream merging and audio conversion.

Installing FFmpeg

- **macOS:** `brew install ffmpeg`
- **Ubuntu/Debian:** `sudo apt install ffmpeg`
- **Windows:** Download from ffmpeg.org/download.html and add the bin/ folder to your system PATH.

VERIFY	Open a terminal and run: ffmpeg -version. You should see version information — if so, FFmpeg is correctly installed.

2.2 Install Python Dependencies

From the unzipped project folder, run:

```
pip install -r requirements.txt
```

Package	Purpose
yt-dlp >= 2024.1.0	Core download engine
customtkinter >= 5.2.0	Modern dark-mode UI framework
Pillow >= 10.0.0	Thumbnail image loading
requests >= 2.31.0	HTTP client for thumbnail fetch
tkinterdnd2 >= 0.3.0	Drag-and-drop support (optional)

NOTE	tkinterdnd2 is optional. If it fails to install, the app works fully — drag-and-drop is simply disabled.

2.3 Launch

```
python main.py
```

On first launch the app checks dependencies and prints instructions for anything missing. The main window opens immediately — no further setup needed.

3. Interface Tour

The main window is divided into five horizontal zones:

3.1 Header Bar

- **App title** — branding on the left.
- ⚙️ **Settings** — opens the Settings dialog.
- 📋 **History** — opens the Download History panel.
- **Dark Mode toggle** — switches between dark and light themes. Preference is remembered.

3.2 URL Input Bar

- **URL field** — paste or type any YouTube video or playlist URL. Press Enter to trigger a fetch.
- 📋 **Paste** — reads from clipboard and fills the URL field.
- 🔍 **Fetch Info** — contacts YouTube and downloads metadata. Disabled during an in-progress fetch.

TIP	Drag a URL from your browser's address bar onto the window — it is pasted and fetched automatically (requires tkinterdnd2).

3.3 Status Bar

Colour-coded state indicator: **blue** = fetching, **amber** = downloading, **green** = complete, **red** = error.

3.4 Video Info & Format Panel

- **Thumbnail** — 240×135 px preview loaded from YouTube.
- **Title, Channel, Duration** — video metadata.
- **Playlist badge** — shown when a playlist URL is detected.
- **Format selector** — MP4 / WebM / MP3 / M4A / Opus / WAV.
- **Quality selector** — resolution for video; disabled (Best) for audio-only formats.
- **Save to folder** — click Browse to change the output directory.
- **Estimated size** — calculated from yt-dlp format metadata when available.
- ⬇ **Download** — starts the download.

3.5 Progress & Log Panel

- **Filename** — current file being written.
- **Speed, ETA, Size** — real-time transfer statistics.
- **Progress bar** — 0–100% fill.
- **Log console** — colour-coded scrolling event log. Click Clear to wipe.

3.6 Control Strip (bottom)

- ⏸ **Pause /** ▶ **Resume** — pauses or resumes the active download.
- ✕ **Cancel** — cancels immediately. Both buttons are disabled when idle.

4. Step-by-Step Usage

4.1 Single Video

1. Copy a YouTube video URL from your browser.
2. Paste it into the URL field and press Enter (or click 🔍 Fetch Info).
3. Wait for the info panel to populate with the title, thumbnail, and duration.
4. Choose Format (e.g. MP4) and Quality (e.g. 1080p).
5. Confirm or change the Save to folder.
6. Click ⬇ Download and monitor the progress bar.
7. When the log shows "Download complete! ✓" the file is in your folder.

4.2 Audio Only

8. Fetch the video as above.
9. Set Format to MP3 (or M4A / Opus / WAV).
10. Quality auto-switches to **Best** and is disabled.
11. Click ⬇ Download. FFmpeg extracts and encodes the audio stream.

NOTE	MP3 is encoded at 192 kbps. For archival quality use M4A (AAC) which avoids re-encoding.

4.3 Playlist

Paste a YouTube playlist URL (contains /playlist?list=...). After fetching:

12. A scrollable checklist shows up to 50 videos — all ticked by default.
13. Untick any videos you want to skip.
14. Choose Format, Quality, and folder as normal.
15. Click ⬇ Download. All ticked videos download sequentially.

TIP	For very large playlists, download in batches of 20–30 for easier progress tracking and recovery from interruptions.

4.4 Pause & Resume

- Click ⏸ **Pause** — suspends the download at the current chunk boundary.
- Click ▶ **Resume** — continues from where it paused in memory.

NOTE	Pause/Resume works within the current session only. Cancelling or closing the app

	discards the partial download.

4.5 Cancel

Click ✕ **Cancel** at any time to stop immediately. Any partial file on disk may remain — it is safe to delete it.

5. Format & Quality Reference

Video Formats

Format	Notes
MP4	Most compatible. Works on all devices, browsers, and media players. Uses H.264 + AAC. Recommended for general use.
WebM	Open-source. Smaller files than MP4 at equivalent quality. Uses VP9 + Opus. Best for web publishing.

Audio Formats

Format	Notes
MP3	Universal compatibility. 192 kbps CBR. Great for music players, sharing. Lossy.
M4A	AAC in MPEG-4. Better quality than MP3 at same bitrate. Preferred for Apple devices.
Opus	Best compression-to-quality ratio. Ideal for speech/podcasts. Less universally supported.
WAV	Uncompressed lossless. Very large files. Use only when further audio editing is planned.

Resolution Options

Resolution	Description
Best	Highest available. Recommended for quality-first downloads.
2160p	4K Ultra HD. ~2–6 GB/hr.
1440p	2K / QHD. ~1–3 GB/hr.
1080p	Full HD. Best practical choice. ~500 MB–1.5 GB/hr.
720p	HD. Good quality, smaller files. ~200–500 MB/hr.
480p	SD. Suitable for small screens. ~100–200 MB/hr.
360p	Low quality. For limited bandwidth.
240p	Very low quality.
144p	Minimum. For testing only.

NOTE	If a requested resolution is unavailable, yt-dlp automatically falls back to the next best

	available option.

6. Settings & History

Settings Dialog

Click ⚙ **Settings** in the header bar. Changes are saved to `~/.ytdownloader/config.json`.

Setting	Description
Embed metadata	Write title, artist, and other tags into the output file. On by default.
Embed thumbnail	Embed cover art in MP4/M4A files. Requires FFmpeg. On by default.
Concurrent downloads	Parallel fragment downloads per file (1–5). Higher = faster on fast connections. Default: 2.

Config File

Settings are stored at `~/.ytdownloader/config.json`. Delete this file to reset all settings to defaults.

Download History

Click 📋 **History** in the header bar to see completed downloads (title + format). The last 200 entries are kept. Click **Clear History** to wipe the log.

7. Keyboard Shortcuts

Shortcut	Action
Enter	In the URL field: triggers Fetch Info.
Ctrl + V	Pastes clipboard content into the URL field.
Drag & Drop	Drag a URL from a browser onto the window to paste and fetch automatically.

8. Updating yt-dlp

YouTube's internal API changes frequently. When downloads suddenly fail, **update yt-dlp first**:

```
pip install -U yt-dlp
```

Only `app/downloader/yt_downloader.py` imports yt-dlp. Updating the library requires no changes to any other file.

TIP	yt-dlp releases updates very frequently. If you see extraction errors, update before anything else.

9. Error Reference

All errors appear in the Log Console with a red label. Common errors and fixes:

Error	Cause & Fix
This video is private	Video is private. Requires login cookies — not supported in the standard install.
Age-restricted video	YouTube requires login verification. Run yt-dlp manually with --cookies-from-browser.
Not available in your region	Video is geo-blocked. Use a VPN and restart.
Unable to extract video info	URL is invalid or not a YouTube URL. Check the link.
Requested format not available	Chosen resolution does not exist for this video. Try a lower resolution or Best.
Network error / timed out	Internet connection problem. Check your connection and retry.
Copyright / removed	Video has been taken down. Nothing can be done.
No information returned	URL points to a live stream or unsupported content type.
ffmpeg not found	FFmpeg is not installed or not on PATH. See Section 2.1.
Cancelled by user	You clicked Cancel. Informational, not an error.

10. Troubleshooting

App does not launch

- **Missing packages:** Re-run `pip install -r requirements.txt`.
- **Wrong Python version:** Must be 3.11+. Check with `python --version`.
- **Tkinter not installed (Linux):** `sudo apt install python3-tk`.

Fetch always fails

- **yt-dlp outdated:** Run `pip install -U yt-dlp`.
- **Invalid URL:** Must start with https://www.youtube.com/ or https://youtu.be/
- **Network issue:** Check your internet connection.

Download completes but no file appears

- **Wrong folder:** Check the Save to path.
- **FFmpeg merge failed:** Confirm with `ffmpeg -version` in a terminal.

Video has no audio (or vice versa)

FFmpeg failed to merge streams. Confirm it is installed (`ffmpeg -version`) and retry.

Slow downloads

- Increase Concurrent downloads in Settings.
- Try switching format (WebM ↔ MP4) — YouTube throttles some formats differently.
- Restarting the download often gives a fresh, unthrottled connection.

11. Project Structure

File	Purpose
`main.py`	Entry point. Checks dependencies, launches MainWindow.
`requirements.txt`	Python package dependencies.
`app/ui/main_window.py`	Root window + controller. Owns YTDownloader and routes callbacks.
`app/ui/download_panel.py`	Thumbnail, video info, format selectors, Download button.
`app/ui/progress_panel.py`	Progress bar, stats, scrolling log console.
`app/downloader/yt_downloader.py`	yt-dlp imported ONLY here. All download logic.
`app/downloader/format_parser.py`	Format dataclasses and yt-dlp selector string builders.
`app/utils/settings.py`	JSON-backed persistent settings. Module-level singleton.
`app/utils/logger.py`	Bridges Python logging to the UI log console.

Design Principles

- **yt-dlp isolation:** Imported in exactly one file. Updating or swapping the engine requires minimal changes.
- **Thread safety:** All downloads run in daemon threads. UI updates scheduled via .after(0, ...) on the main thread.
- **Event-driven:** yt-dlp hooks emit DownloadProgress dataclasses. No polling, no shared mutable state.
- **Settings as source of truth:** All persistent state goes through the settings singleton.
- **Error isolation:** All yt-dlp exceptions converted to friendly messages before reaching the UI.

12. Legal & Responsible Use

IMPORTANT	This software is provided as a technical tool. You are solely responsible for ensuring your use complies with applicable laws and platform terms of service.

- Downloading videos for personal offline viewing may be permitted in some jurisdictions but is prohibited by YouTube's Terms of Service.
- Do not download and redistribute copyrighted content without permission from the rights holder.
- Creative Commons and public-domain content may generally be downloaded — check the specific licence.
- This tool is intended for legal uses: archiving your own content, downloading CC-licensed material, and educational research.

The developers of YT Downloader accept no liability for misuse. Use responsibly and respect content creators.

YT Downloader v1.0 · Built with yt-dlp, CustomTkinter & Python

https://github.com/your-repo/ytdownloader